

The original current, its source overlap and the collision resolvent

Split-Zero research programme

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The same two complex coordinates determine the marked arithmetic current and the next-source overlap. We calculate their joint feasible region, including the retained coordinate cap, and the sharp bound for every real linear combination needed by the native source step. The measured-mass interval is handled without assuming monotonicity. All metric maps and equality cases are included.

The second calculation retains the full arithmetic operator and controls its change at the observed dual-number collision on the stated exterior frequency region. Its error is bounded by the original source leakage and area data at all four cutoffs. The interior native phase and its particular marked sign remain unevaluated. This source includes complete proofs and the unchanged preceding proof collection.

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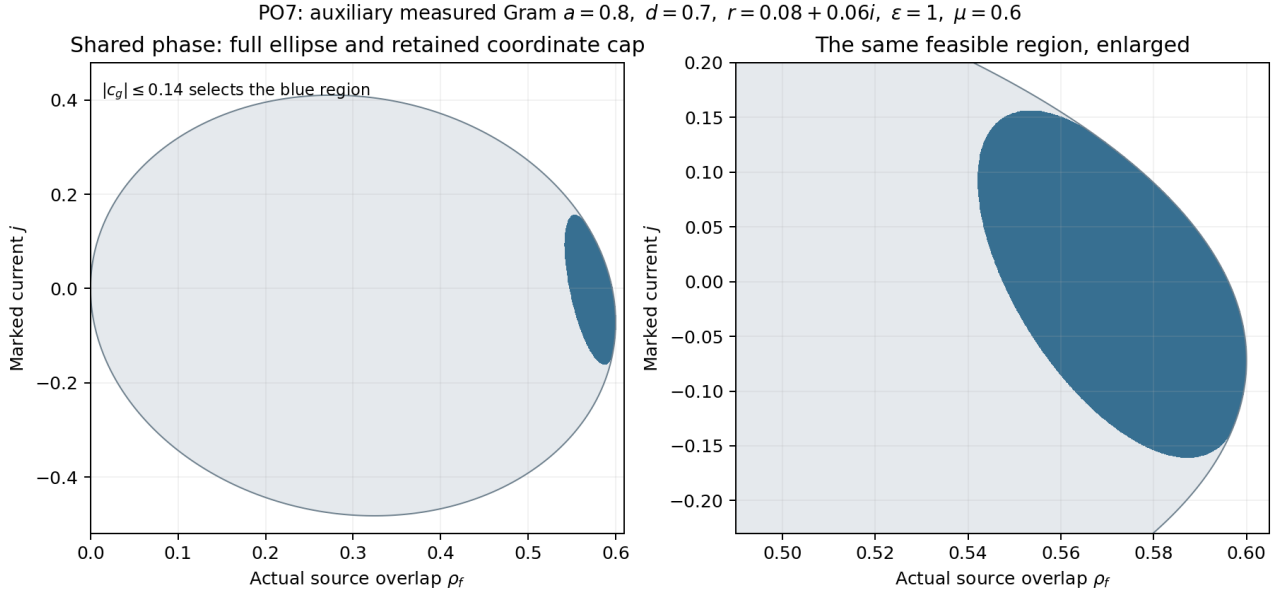


Figure 1: PO4–8 gives an exact coordinate map and reconstruction of the shared phase. The grey ellipse drops the coordinate cap; the blue region retains it. The right panel enlarges the same region. These displayed positive Gram data are auxiliary probes, with all constants printed above the figure. They are not evaluated native periods or zeta zeros. PO9–18 optimizes a common phase in this region instead of combining two independently chosen extrema.

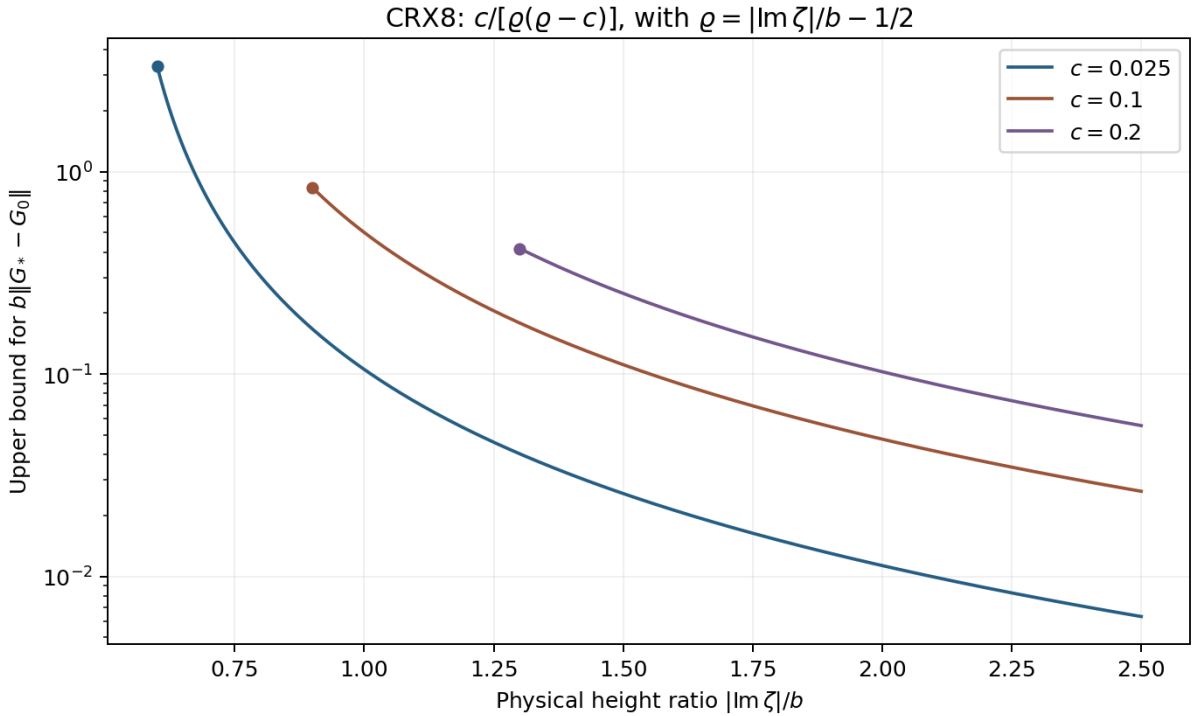


Figure 2: CRX7–8 proves the plotted upper bounds for the full original resolvent, including its selfadjoint part. Here $b = \epsilon\sqrt{D}$ and $c = |r|/\sqrt{D}$. Each curve starts at its exact sufficient margin $\varrho = 4c$. The physical frequency is related to the horizontal coordinate by the displayed ratio; the figure does not identify different cutoff scales. The curves are proved bounds at the stated auxiliary values of c , not sampled native resolvent norms. CRX10–12 retains the logarithmic determinant and all four original cutoff errors.

The exact joint region of original current and native source overlap

22 September 2026. Independent derivation PO1-PO20. The native source-step curvature contains the overlap with the same second boundary vector that also enters the marked current. Their phases therefore cannot be optimized independently. This proof gives their complete feasible region, sharp combined envelopes, every degenerate case, and the exact receiver into the actual adjacent source step. No phase or sign is assigned to the marked terminal class.

1. The original objects and the retained phase

At one literal original cutoff N , retain the complete source metric G_N , observation Λ , minimum section L_N , and isometry

$$Q_N = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L_N = G_N^{-1} \Lambda^* Q_N, \quad J = L_N Q_N^{-1/2}.$$

The original multiplication operator and boundary vectors obey

$$M = C + \epsilon f e^\dagger, \quad C = C^\dagger, \quad e^\dagger f = 0, \quad \|e\| = \|f\| = 1, \\ \epsilon > 0.$$

These are the exact OCP5-7 source classes. All daggers before observation use G_N .

Put

$$u = J^\dagger e, \quad v = J^\dagger f, \quad a = \|u\|^2, \quad d = \|v\|^2, \quad r = u^* v = x_r + ib, \\ D = ad - |r|^2 > 0.$$

$$g = u/\sqrt{a}, \quad h = (v - (r/a)u)/\sqrt{D/a}, \quad W = i\epsilon(vu^* - uv^*). \quad (\text{PO1})$$

The strict positivity is the original complete two-column area certificate; it is not inferred from $a > 0$ alone. The vectors g, h are orthonormal in the actual observed metric coordinates. The current vanishes on their orthogonal complement and has matrix

$$W|_{g,h} = \begin{pmatrix} -2\epsilon b & -i\epsilon\sqrt{D} \\ i\epsilon\sqrt{D} & 0 \end{pmatrix}. \quad (\text{PO2})$$

For the original observed terminal unit vector $\hat{y}$, or a probe being compared under the same finite constraints, write

$$c_g = g^* \hat{y}, \quad c_h = h^* \hat{y}, \quad \mu = |c_g|^2 + |c_h|^2, \quad |c_g| \leq \eta, \\ 0 \leq \mu \leq 1.$$

The actual terminal $\eta = \eta_N$ is the bound in OSP38-43, carried from TR10 without changing its domain. Define

$$j = \hat{y}^* W \hat{y} = 2\epsilon\sqrt{D} \Im(\bar{c}_g c_h) - 2\epsilon b |c_g|^2, \\ \rho = \rho_f = \frac{|v^* \hat{y}|^2}{d}. \quad (\text{PO3})$$

Here ρ is the actual native second-source overlap. It is not a free parameter independent of j .

2. An exact joint phase/overlap region

Use the orthonormal basis of the same current plane

$$v_0 = v/\sqrt{d} = \frac{rg + \sqrt{D}h}{\sqrt{ad}}, \quad n_0 = \frac{\sqrt{D}g - \bar{r}h}{\sqrt{ad}}. \quad (\text{PO4})$$

Direct inner products prove orthogonality and unit norms, including every complex conjugation. Put

$$A = v_0^* \hat{y}, \quad B = n_0^* \hat{y}, \quad |A|^2 = \rho, \quad |B|^2 = \mu - \rho, \\ \bar{A}B = X + iY.$$

Multiplying PO2 by the two columns in PO4 gives

$$W|_{v_0, n_0} = \begin{pmatrix} -2\epsilon b & i\epsilon\sqrt{D} \\ -i\epsilon\sqrt{D} & 0 \end{pmatrix}.$$

Consequently

$$j = -2\epsilon b\rho - 2\epsilon\sqrt{D}Y, \quad X^2 + Y^2 = \rho(\mu - \rho). \quad (\text{PO5})$$

This is the exact shared phase, not only a Cauchy bound.

For any proposed real pair (j, ρ) , define

$$Y(j, \rho) = -\frac{j + 2\epsilon b\rho}{2\epsilon\sqrt{D}}, \\ R(j, \rho) = \rho(\mu - \rho) - Y(j, \rho)^2, \\ Z_{\text{mid}}(j, \rho) = \frac{D\mu + (x_r^2 - b^2 - D)\rho - (b/\epsilon)j}{ad}, \\ Z_{\text{min}}(j, \rho) = Z_{\text{mid}}(j, \rho) - \frac{2\sqrt{D}|x_r|}{ad}\sqrt{R(j, \rho)} \\ \text{when } R(j, \rho) \geq 0. \quad (\text{PO6})$$

Then the complete feasible region for the mass and coordinate constraints is

$$\boxed{0 \leq \rho \leq \mu, \quad R(j, \rho) \geq 0, \quad Z_{\text{min}}(j, \rho) \leq \eta^2.} \quad (\text{PO7})$$

All three conditions are necessary and sufficient.

Here is the full reconstruction proof. Invert PO4 on its plane to obtain $c_g = (rA + \sqrt{D}B)/\sqrt{ad}$. Therefore

$$|c_g|^2 = \frac{D\mu + (|r|^2 - D)\rho + 2\sqrt{D}(x_r X + bY)}{ad} \\ = Z_{\text{mid}} + \frac{2\sqrt{D}x_r}{ad}X.$$

PO5 requires $X = \pm\sqrt{R}$; these two possibilities give exactly the two values $Z_{\text{mid}} \pm 2\sqrt{D}|x_r|\sqrt{R}/(ad)$. If PO7 holds and $\rho > 0$, choose $A = \sqrt{\rho}$ and $B = (X + iY)/\sqrt{\rho}$, using the sign of X which gives Z_{min} . They have the required norms and current, and $|c_g|^2 = Z_{\text{min}}$. The reconstructed value automatically lies between zero and μ , since PO4 is unitary. Thus no additional lower bound for Z_{min} has been silently dropped.

If $\rho = 0$, PO7 forces $Y = j = 0$ and $R = 0$; choose $A = 0$, $B = \sqrt{\mu}$. If $\rho = \mu$, take $B = 0$, $A = \sqrt{\mu}$; the current is then $j = -2\epsilon b\mu$. If $\mu = 0$, the only feasible pair is $(0, 0)$. These choices prove sufficiency at every endpoint as well.

To obtain an observed unit vector when $\mu < 1$, add a vector of norm $\sqrt{1-\mu}$ in the orthogonal complement of the current plane. This complement is nonzero in the actual programme: OCS32's five-orbit rank bound gives $\dim B \geq \lfloor q/5 \rfloor + \mathbf{1}_{5|k} > 2$. In a separate two-dimensional model unit vectors require $\mu = 1$; the phase formulas still hold for vectors of squared norm μ .

Two useful degenerate geometries follow without dividing by x_r :

$$x_r = 0 : \quad |c_g|^2 = Z_{\text{mid}} \quad \text{for both allowed signs of } X;$$

$$r = 0 : \quad \boxed{\max\{0, \mu - \eta^2\} \leq \rho \leq \mu, \quad j^2 \leq 4\epsilon^2 D\rho(\mu - \rho)}. \quad (\text{PO8})$$

For general r , dropping only the coordinate cap gives the exact filled ellipse $(j + 2\epsilon b\rho)^2 \leq 4\epsilon^2 D\rho(\mu - \rho)$. With the cap, PO7 retains the necessary extra restriction. It does not replace the actual region by an independently chosen interval for each variable.

3. Sharp current-plus-overlap envelopes

For any real coefficient θ , the same phase-dependent expression is

$$j + \theta\rho = \hat{y}^*(W + \theta P_v)\hat{y}, \quad P_v = vv^*/d.$$

In the original g, h frame write

$$\begin{aligned} W + \theta P_v &= \begin{pmatrix} A_\theta & H_\theta \\ \bar{H}_\theta & B_\theta \end{pmatrix}, \\ A_\theta &= -2\epsilon b + \frac{\theta|r|^2}{ad}, \quad B_\theta = \frac{\theta D}{ad}, \\ H_\theta &= \sqrt{D} \left(\frac{\theta r}{ad} - i\epsilon \right), \\ \delta_\theta &= A_\theta - B_\theta. \end{aligned} \quad (\text{PO9})$$

All phases remain. Exact expansion gives

$$\begin{aligned} \det(W + \theta P_v) &= -\epsilon^2 D, \quad \Gamma_\theta = \sqrt{\delta_\theta^2 + 4|H_\theta|^2} \\ &= \sqrt{(\theta - 2\epsilon b)^2 + 4\epsilon^2 D} > 0. \end{aligned} \quad (\text{PO10})$$

Thus this entire auxiliary Hermitian pencil remains indefinite on the current plane, even though its restriction to a narrow coordinate cap can have one sign. The determinant cancellation follows because $A_\theta B_\theta$ and $|H_\theta|^2$ have identical quadratic and mixed θ terms.

For $\sigma \in \{+1, -1\}$, define

$$\begin{aligned} z_\sigma(\theta, \mu) &= \min \left\{ \eta^2, \frac{\mu}{2} \left(1 + \frac{\sigma \delta_\theta}{\Gamma_\theta} \right) \right\}, \\ \mathcal{M}_\sigma(\theta, \mu) &= \sigma B_\theta \mu + \sigma \delta_\theta z_\sigma \\ &\quad + 2|H_\theta| \sqrt{z_\sigma(\mu - z_\sigma)}. \end{aligned} \quad (\text{PO11})$$

The minimum is nonnegative and at most μ . The exact sharp bounds are

$$\boxed{-\mathcal{M}_-(\theta, \mu) \leq j + \theta\rho \leq \mathcal{M}_+(\theta, \mu)}. \quad (\text{PO12})$$

The numbers $\mathcal{M}_\sigma$ need not be positive under a severe coordinate cap; they are the actual extrema of $\sigma(j + \theta\rho)$, with their signs retained.

To prove PO12 put $z = |c_g|^2$, so $0 \leq z \leq \min(\eta^2, \mu)$. For this fixed z , the maximum of $\sigma(j + \theta\rho)$ is exactly

$$\sigma B_\theta \mu + \sigma \delta_\theta z + 2|H_\theta| \sqrt{z(\mu - z)}. \quad (\text{PO13})$$

If $H_\theta \neq 0$, equality is achieved by

$$c_g = \sqrt{z}, \quad c_h = \sigma \frac{\overline{H}_\theta}{|H_\theta|} \sqrt{\mu - z}.$$

For $\mu > 0$, PO13 is strictly concave on $0 < z < \mu$, and its derivative vanishes uniquely at $\mu(1 + \sigma\delta_\theta/\Gamma_\theta)/2$. Indeed the derivative is $\sigma\delta_\theta + |H_\theta|(\mu - 2z)/\sqrt{z(\mu - z)}$; substitution gives zero, with the sign fixed before squaring. Truncating this maximizer by the cap proves PO11.

If $H_\theta = 0$, PO13 is affine in z . By PO10, $\delta_\theta \neq 0$; hence its maximum is at $z = 0$ or the cap endpoint according to the sign of $\sigma\delta_\theta$, exactly as PO11 states. This degeneracy can occur only when $x_r = 0$, $b \neq 0$, and $\theta = \epsilon ad/b$. If $\mu = 0$, every expression is zero. If the chosen z is zero or μ , the vanishing amplitude makes its relative phase immaterial. These cases cover every equality and degeneracy scope.

At $\theta = 0$, one has $\delta_0 = -2\epsilon b$, $|H_0| = \epsilon\sqrt{D}$, $\Gamma_0 = 2\epsilon\sqrt{b^2 + D}$. PO11-12 therefore reduce exactly to OSP48-50, preserving the signed rank-one correction. For $\theta \neq 0$, their common phase is optimized in PO13; adding independent optimal bounds for j and ρ generally cannot attain this value.

4. Complete measured-mass interval receiver

The original measured-support estimate supplies $m_- \leq \mu \leq m_+$, with

$$m_- = \max\{0, (\mathfrak{s} - E^{\text{supp}})/d\}, \quad m_+ = \min\{1, (\mathfrak{s} + E^{\text{supp}})/d\}. \quad (\text{PO14})$$

The proof and original errors are OSP39-41. For a general θ , replacing μ by m_+ in PO12 is not always valid: the maximum can decrease with mass on a narrow coordinate cap.

There is nevertheless an exact finite calculation using at most three candidates. Put $h_\theta = |H_\theta|$ and $Z = \eta^2$. For each sign σ , start with the two candidates $\mathcal{M}_\sigma(\theta, m_-)$ and $\mathcal{M}_\sigma(\theta, m_+)$. When $0 < Z \leq m_+$, put

$$v_{\min} = \max\{0, m_- - Z\}, \quad v_{\max} = m_+ - Z.$$

On the active cap $z = Z$, define

$$f_{\sigma, Z}(v) = \sigma A_\theta Z + \sigma B_\theta v + 2h_\theta \sqrt{Zv}.$$

If $\sigma B_\theta < 0$ and $h_\theta > 0$, set

$$v_* = \min\{v_{\max}, \max\{v_{\min}, h_\theta^2 Z / B_\theta^2\}\}.$$

Otherwise take the endpoint of $[v_{\min}, v_{\max}]$ maximizing this displayed scalar function. This latter case is monotone when $h_\theta > 0$ and affine when $h_\theta = 0$, so its endpoint evaluation is exact. Include $f_{\sigma, Z}(v_*)$ as the third candidate. Let $\mathcal{U}_\sigma(\theta)$ be the largest included candidate. Then

$$\boxed{\begin{array}{c} \sup \\ m_- \leq \mu \leq m_+ \\ |c_g|^2 \leq \eta^2 \\ \sigma(j + \theta\rho) = \mathcal{U}_\sigma(\theta). \end{array}} \quad (\text{PO15})$$

The supremum is over the precise mass/coordinate information; the remaining observed norm is supplied as in PO7.

Here is a proof that no interior optimizer is missed. After optimizing the phase, put $z = |c_g|^2$, $v = |c_h|^2$. The feasible set is $z, v \geq 0$, $z \leq Z$, $m_- \leq z + v \leq m_+$. The objective is $\sigma A_\theta z + \sigma B_\theta v + 2h_\theta \sqrt{zv}$. At a differentiable interior stationary point its two partial derivatives would give $A_\theta B_\theta = h_\theta^2$, contradicting the strict negative determinant PO10. Thus a maximum lies on a boundary. The two mass boundaries give PO11. The boundary $z = Z$ gives the scalar calculation above. At $z = 0$ or $v = 0$ the function is linear in the other coordinate, so an endpoint lies on one of those previously listed boundaries. If $Z = 0$, the two mass endpoints already suffice. If $Z > m_+$, that cap cannot bind. If $h_\theta = 0$, the entire objective is affine and its vertices are among the same candidates. Compactness proves attainment.

This prevents an unsupported monotonicity step when the source trace correction has either sign.

5. Receiver into the actual adjacent-source current

Use the native adjacent step and notation of NV1-26, keeping the two literal paths $q - 1 \rightarrow q$ and $2q - 1 \rightarrow 2q$. Let $Y_0 = \xi^* Q_N \xi > 0$, $\beta_{\text{src}} = z_1^* Q_N z_1 / \omega_{N+1}$, and $\Delta\tau = \tau_{N+1} - \tau_N$. The same source vector satisfies $w = \beta_{\text{src}} Y_0 \rho$, with precisely the overlap in PO3. For $0 \leq t \leq 1$, put

$$\lambda = \frac{(1 + \beta_{\text{src}})t}{1 + \beta_{\text{src}}t}, \quad \theta_\lambda = \lambda \frac{\beta_{\text{src}}}{1 + \beta_{\text{src}}} \Delta\tau. \quad (\text{PO16})$$

The exact source-step formula NV8-9 now groups the correlated quantities:

$$\boxed{\frac{\Phi(t)}{Y_0} = (1 - \lambda)[j + \theta_\lambda \rho] + \lambda \frac{\Phi_1}{Y_0}.} \quad (\text{PO17})$$

Here $j = \Phi_0/Y_0$ is the endpoint current of the same normalized observed vector. The trace difference and source parameter are the actual ones, not chosen to force a sign. Substituting PO12 yields

$$\boxed{\begin{aligned} \lambda \frac{\Phi_1}{Y_0} - (1 - \lambda) \mathcal{M}_-(\theta_\lambda, \mu) \\ \leq \frac{\Phi(t)}{Y_0} \leq \\ \lambda \frac{\Phi_1}{Y_0} + (1 - \lambda) \mathcal{M}_+(\theta_\lambda, \mu). \end{aligned}} \quad (\text{PO18})$$

If only PO14's mass interval is available, replace $\mathcal{M}_\sigma$ by $\mathcal{U}_\sigma$ from PO15. At $\lambda = 1$, the prefactor is zero and the exact endpoint Φ_1 remains; no limiting division by $1 - \lambda$ is used. At $\lambda = 0$, the original OSP bound is recovered. These envelopes are sharp for the base mass/coordinate-constrained expression $j + \theta\rho$. Requiring a specified second-endpoint value can impose further correlations, so no independent attainability of all full-path endpoint data is claimed.

The exact doubled-rate mass receiver also remains useful:

$$\begin{aligned} 0 \leq \mu - \rho &= \frac{|\sqrt{D} c_g - r c_h|^2}{ad} \\ &\leq \frac{(\sqrt{D} \eta + |r|)^2}{ad} =: E_f. \end{aligned}$$

Consequently

$$\boxed{|(j + \theta\rho) - (j + \theta\mu)| \leq |\theta| E_f.} \quad (\text{PO19})$$

For the actual terminal class, NV16-19 gives $E_f \leq e^{-2q\psi(s_N)+O_{h,\omega}(k \log q)}$. On either native adjacent path, $|\Delta\tau| \leq 2\epsilon_N\sqrt{\alpha_N} + 2\epsilon_{N+1}\sqrt{\alpha_{N+1}} = e^{O(k \log q)}$, so PO19 with θ_λ has the doubled rate uniformly over the whole source step. This is an absolute error bound; it makes no division by μ or by a signed current.

Every statement is applied separately at all four original cutoffs. If $F_N = j_N + \theta_N \rho_N$ and PO12 gives $-M_{-,N} \leq F_N \leq M_{+,N}$, the signed return with the original $+, +, -, -$ signs lies within the following explicit interval

$$\begin{aligned} & - \sum_{N \in \{q-1, q\}} M_{-,N} - \sum_{N \in \{2q-1, 2q\}} M_{+,N} \\ & \leq \mathcal{R}F_N \leq \sum_{N \in \{q-1, q\}} M_{+,N} \\ & \quad + \sum_{N \in \{2q-1, 2q\}} M_{-,N}. \end{aligned} \tag{PO20}$$

The triangle/interval addition uses the four actual values; it does not identify their phases.

6. Source and verification scope

OSP38-51 was read in full in the published node source linked above. Its exact current correction and fixed-mass optimizer are recovered at $\theta = 0$. The new results here are the exact joint feasible region PO7, the determinant-preserving combined pencil and capped envelopes PO9-13, the complete mass-interval calculation PO15, and the common-phase native receiver PO17-19.

The exact finite checks verify all unitary coordinate changes, complex signs, determinant cancellation, stationary points, degenerate off-diagonal cases, the two possible joint-region reconstructions, and source-step regrouping. Equality examples are probes with the stated source geometry and mass constraints, not evaluations of an actual zeta zero or assertions that the original terminal probe realizes an extremum. The source's human Gamma and elliptic citations remain those in the cited OCP/RC/OSP derivations; all additional finite phase geometry and optimization claims are proved in full here.

The collision comparison for the full original arithmetic resolvent

22 September 2026. This calculation applies the measured collision data to the full observed arithmetic operator. It preserves the selfadjoint part, the observation metric, the actual spectral coordinate and all four source cutoffs. The new estimate concerns exterior spectral coordinates; it does not evaluate the native phase integral on its remaining frequency region.

1. Original objects and source identities

At a literal original cutoff $N \in \{q-1, q, 2q-1, 2q\}$, retain the positive source metric G_N , onto observation Λ , and minimum isometry

$$Q_B = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad L = G_N^{-1} \Lambda^* Q_B, \quad J = L Q_B^{-1/2}. \quad (\text{CRX1})$$

A dagger on the source uses G_N ; a star on the observed space uses the Euclidean coordinates given by this exact isometry. In the original decomposition $M = C + \epsilon f e^\dagger$, retain $C = C^\dagger$, the orthonormal source vectors e, f , and $\epsilon > 0$. Put

$$\begin{aligned} u &= J^\dagger e, \quad v = J^\dagger f, \quad a = u^* u, \quad d = v^* v, \quad r = u^* v, \\ D &= ad - |r|^2 \geq \Theta_N > 0, \quad \alpha = 1 - a. \end{aligned} \quad (\text{CRX2})$$

These are the original ST5-6 and ST13-20 data, with their finite native guards. The actual terminal-cardinal estimates retain the five-orbit and admitted-period domain specified there. The finite matrix proof below uses the displayed positive-area data and applies throughout that established domain.

Write

$$C_B = J^\dagger C J = C_B^*, \quad M_B(s) = C_B + (\epsilon v + s u) u^*, \quad s_* = -\epsilon r / a,$$

$$\begin{aligned} b &= \epsilon \sqrt{D} > 0, \quad c = \frac{|r|}{\sqrt{D}}, \quad \beta_r = \Im r, \\ M_\theta &= M_B(\theta s_*), \quad 0 \leq \theta \leq 1. \end{aligned} \quad (\text{CRX3})$$

The symbol β_r denotes the imaginary part of the retained cross-Gram entry. It is not the source leverage called β in NV. The complex parameter s , straight-segment parameter θ , and spectral coordinate ζ have separate roles. The underlying metric is fixed at this particular cutoff throughout the s -deformation.

The exact collision and rank-one determinant/inverse identities are OC1-17. The complete local proof source was read, together with OSP1-51. CRX4-12 prove the quantitative consequence, rather than assuming a resolvent estimate for the nonnormal full operator.

2. A singular-value bound from the actual current

In the original orthonormal observed frame $g = u/\sqrt{a}$, $h = (v - (r/a)u)/\sqrt{D/a}$, direct multiplication gives

$$\begin{aligned} W_\theta &= i(M_\theta - M_\theta^*) \\ &= \begin{pmatrix} -2\epsilon(1-\theta)\beta_r & -ib \\ ib & 0 \end{pmatrix} \\ &\quad \text{on } \text{span}\{g, h\}, \end{aligned} \quad (\text{CRX4})$$

and zero on its orthogonal complement. The same C_B remains in every full M_θ ; its contribution to this displayed difference cancels by selfadjointness. The two eigenvalues of CRX4 are $-x_\theta \pm \sqrt{b^2 + x_\theta^2}$, where $x_\theta = \epsilon(1 - \theta)\beta_r$. Thus

$$\begin{aligned} h_\theta &:= \tfrac{1}{2}\|W_\theta\| = \tfrac{1}{2}\left(\sqrt{b^2 + x_\theta^2} + |x_\theta|\right) \\ &\leq b/2 + \epsilon|\beta_r| \leq b(1/2 + c), \quad h_1 = b/2. \end{aligned} \tag{CRX5}$$

The first inequality follows from $\sqrt{b^2 + x^2} \leq b + |x|$, proved by squaring the nonnegative sides.

For every unit observed vector x ,

$$\Im\langle x, (\zeta I - M_\theta)x \rangle = \Im\zeta + \tfrac{1}{2}\langle x, W_\theta x \rangle.$$

Cauchy-Schwarz and the triangle inequality imply

$$\|(\zeta I - M_\theta)x\| \geq |\Im\zeta| - h_\theta.$$

If this right side is positive, the square finite matrix is injective and therefore invertible. Taking the infimum over all unit x proves

$$\boxed{\|G_\theta(\zeta)\| \leq \frac{1}{|\Im\zeta| - h_\theta}, \quad G_\theta = (\zeta I - M_\theta)^{-1}.} \tag{CRX6}$$

The estimate holds for every real part of ζ . It uses neither normality nor a norm bound on C_B . In particular a large or noncommuting C_B has not been deleted from the inverse.

3. Evaluated collision error and marked resolvent receivers

Retain a positive exterior margin $\varrho > 0$ in the original spectral units:

$$|\Im\zeta| \geq (1/2 + \varrho)b, \quad c \leq \varrho/4. \tag{CRX7}$$

Then CRX6 applies to the whole segment. The exact inverse identity, obtained by multiplying both inverses into their defining matrices, is

$$G_1 - G_0 = s_* G_1 u u^* G_0.$$

Since $|s_*| \|u u^*\| = \epsilon|r| = bc$, CRX5-7 give

$$\boxed{b\|G_1 - G_0\| \leq \frac{c}{\varrho(\varrho - c)} \leq \frac{4c}{3\varrho^2}.} \tag{CRX8}$$

The collision inverse has denominator at least $b\varrho$; the native inverse has denominator at least $b(\varrho - c)$. Both denominators, and the physical spectral coordinate, remain in the first bound. For any two original marked vectors x, y , CRX8 gives the explicit receiver

$$|x^*(G_1 - G_0)y| \leq \|x\| \|y\| \frac{c}{b\varrho(\varrho - c)}. \tag{CRX9}$$

There is no identification of these two vector phases. The sharper exact ordered-pair identity in OC15 remains applicable when the two actual scalar factors are available.

Let $p_\theta(\zeta) = \det(\zeta I - M_\theta)$, and put $m_0 = u^* G_0 u$. Multilinearity in the columns gives $p_\theta/p_0 = 1 - \theta s_* m_0$: all terms taking two rank-one perturbation columns vanish. The actual scalar satisfies

$$|s_* m_0| \leq \epsilon|r| \|G_0\| \leq \frac{c}{\varrho - c} \leq \frac{1}{3}.$$

The logarithm based at $\theta = 0$ is therefore the absolutely convergent series $-\sum_{j \geq 1} (\theta s_* m_0)^j / j$ along this entire original deformation segment. It has no winding ambiguity or determinant zero there. Summing absolute values and using $1/j \leq 1$ proves

$$\boxed{\begin{aligned} & \left| \text{Log} \frac{p_1(\zeta)}{p_0(\zeta)} \right| \\ & \leq \frac{c}{\varrho - 2c} \leq \frac{2c}{\varrho}. \end{aligned}} \quad (\text{CRX10})$$

Both the real logarithmic-volume error and the continuously followed determinant-phase error obey CRX10. This is a quantitative evaluation of the collision transfer error; it is not an evaluation of either determinant separately.

4. Native source bounds and all four cutoffs

The contraction $0 \prec U^*U \preceq I_2$ implies $|r|^2 \leq (1-a)(1-d) \leq \alpha$, by the determinant of its positive semidefinite complement. Consequently

$$\boxed{c_N \leq \sqrt{\alpha_N / \Theta_N}.} \quad (\text{CRX11})$$

This gives a finite sufficient check for CRX7: require $\sqrt{\alpha_N / \Theta_N} \leq \varrho_N / 4$, using the already defined original finite quantities. The retained native estimates $\alpha_N \leq \exp[-2q\psi((N+1-q)/q) + O_{h,\varpi}(k \log(q+2))]$ and $-\log \Theta_N = O_{h,\varpi}(k \log(q+2))$ yield the evaluated transfer rate $c_N \leq \exp[-q\psi((N+1-q)/q) + O_{h,\varpi}(k \log(q+2))]$. The effective bounds are CRX8-11 themselves; this asymptotic does not replace any finite guard.

At the four actual cutoffs choose exterior coordinates ζ_N satisfying CRX7 separately, or use a single physical ζ whose imaginary part meets all four inequalities. Define the original signed return on these scalar logarithmic transfers by $\mathcal{R}f = f_{q-1} + f_q - f_{2q-1} - f_{2q}$. Then

$$\boxed{\begin{aligned} & \left| \mathcal{R} \text{Log} \frac{p_{*,N}(\zeta_N)}{p_{0,N}(\zeta_N)} \right| \\ & \leq \sum_{N \in \{q-1, q, 2q-1, 2q\}} \frac{c_N}{\varrho_N - 2c_N} \\ & \leq \sum_N \frac{2}{\varrho_N} \sqrt{\frac{\alpha_N}{\Theta_N}}. \end{aligned}} \quad (\text{CRX12})$$

This follows by applying the four signs and then the triangle inequality. No pair of cutoff metrics, source columns, or source functions has been identified. Each $p_{*,N}$ uses its own collision $s_{*,N}$ and keeps its entire original $C_{B,N}$.

The guard in CRX7 specifies exactly the exterior region proved here. Frequencies failing that guard remain a separate part of the original phase calculation. In particular CRX12 is not integrated over an unbounded interval with a frequency-independent error, and the original interior frequency contribution is not discarded. The calculation establishes an exponentially controlled full-operator collision transfer on its stated exterior region. The actual native marked phase remains an arithmetic datum.

5. Provenance and reproducibility

OC1-17 and OSP1-51 were read in full. ST supplies the original source and metric bounds. The new matrix inequalities, determinant identity and logarithm estimate are proved above. No historical novelty of these linear-algebra operations is claimed. Edition 029's complete source collection retains the predecessor definitions, proofs, authorship and human citations.